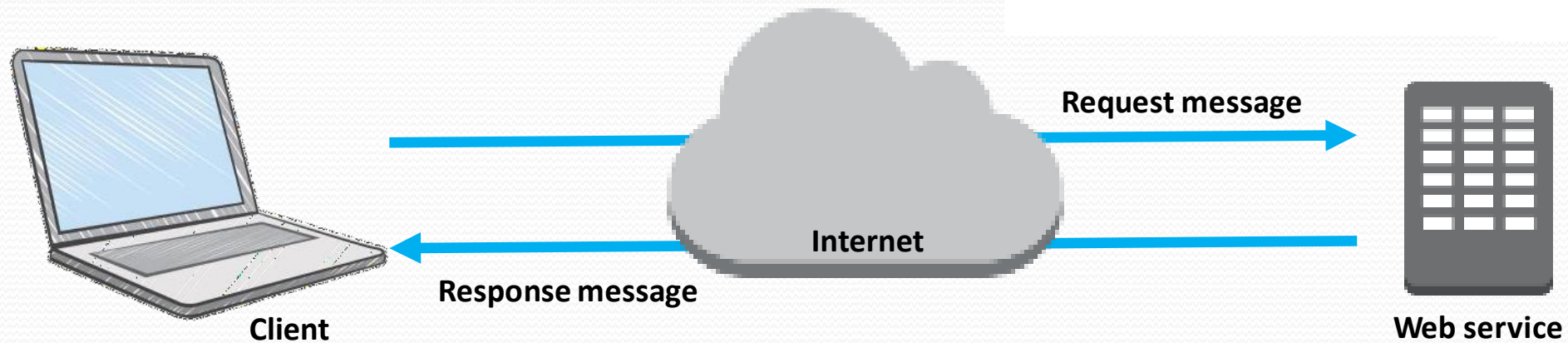
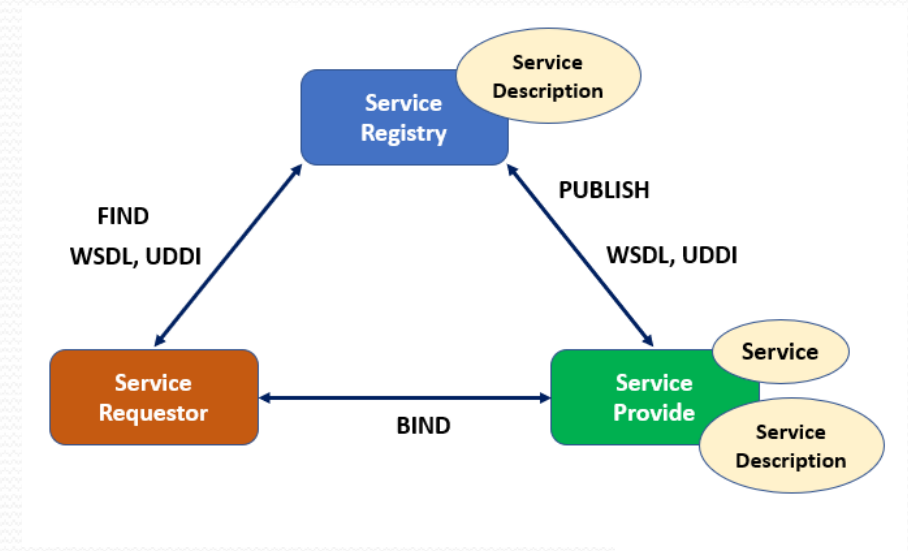


AWS

Web Services

A **web service** is any piece of software that makes itself available over the internet and uses a **standardized format** (XML or JSON) for the request and the response of an **API interaction**.



Existing cloud infrastructure

- The cloud computing infrastructure at Amazon, Google, and Microsoft (as of mid 2012).
 - Amazon is a pioneer in Infrastructure-as-a-Service (IaaS).
 - Google's efforts are focused on Software-as-a-Service (SaaS) and Platform-as-a-Service (PaaS).
 - Microsoft is involved in PaaS.
 - Private clouds are an alternative to public clouds. Open-source cloud computing platforms such as:
 - Eucalyptus,
 - OpenNebula,
 - Nimbus,
 - OpenStack
- can be used as a control infrastructure for a private cloud.

Amazon Web Services (AWS)

- AWS is IaaS cloud computing services launched in 2006.
- Businesses in 200 countries used AWS in 2012.
- The infrastructure consists of compute and storage servers interconnected by high-speed networks and supports a set of services.
- An application developer:
 - Installs applications on a platform of his/her choice.
 - Manages resources allocated by Amazon.

AWS instances

- An instance is a virtual server with a well specified set of resources including: CPU cycles, main memory, secondary storage, communication and I/O bandwidth.
- The user chooses:
 - The region and the availability zone where this virtual server should be placed.
 - An instance type from a limited menu of instance types.
- When launched, an instance is provided with a DNS name; this name maps to a
 - *private IP address* [?] for internal communication within the internal EC2 communication network.
 - *public IP address* [?] for communication outside the internal Amazon network, e.g., for communication with the user that launched the instance.

AWS instances (cont'd)

- Network Address Translation (NAT) maps external IP addresses to internal ones.
- The public IP address is assigned for the lifetime of an instance.
- An instance can request an *elastic IP address*, rather than a public IP address. The elastic IP address is a static public IP address allocated to an instance from the available pool of the availability zone.
- An elastic IP address is not released when the instance is stopped or terminated and must be released when no longer needed.

Amazon Web Services (AWS)



- Amazon reserved instances access to server SSH(linux) and remote desktop.
- Elastic Compute Cloud (EC2) –Xen-based Virtual machines and CPU cycles
- Simple Storage Service (S3) –Virtual storage service
- Simple Queue Service (SQS) –Message passing API
- SimpleDB–Running queries on structured data in real time –works with EC2 and S3
- RDS- Relational Database
- Cloudfront- content delivery
- Cloud watch – Auto scaling & Elastic Load balancing features.

Amazon IaaS Cloud

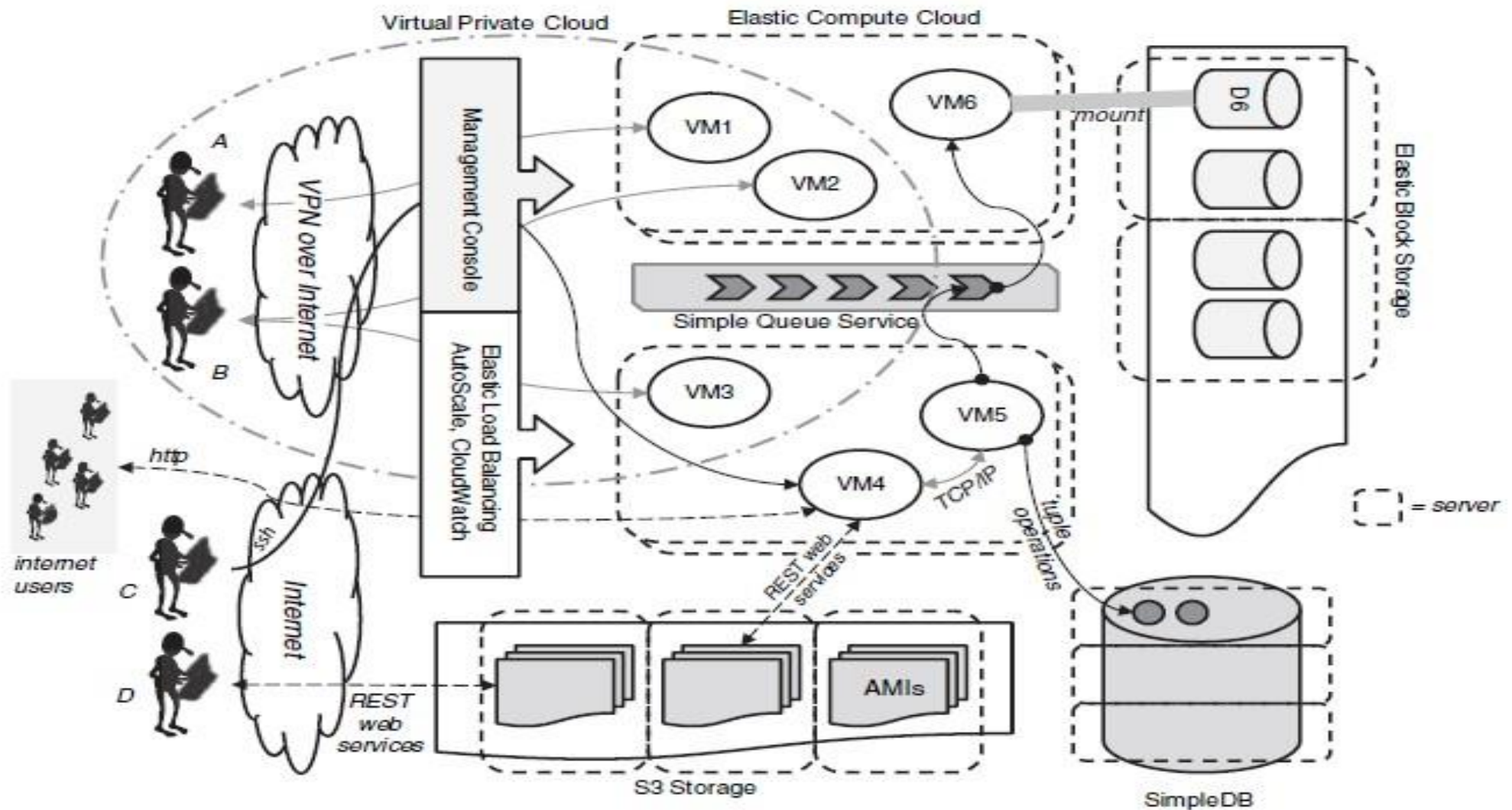
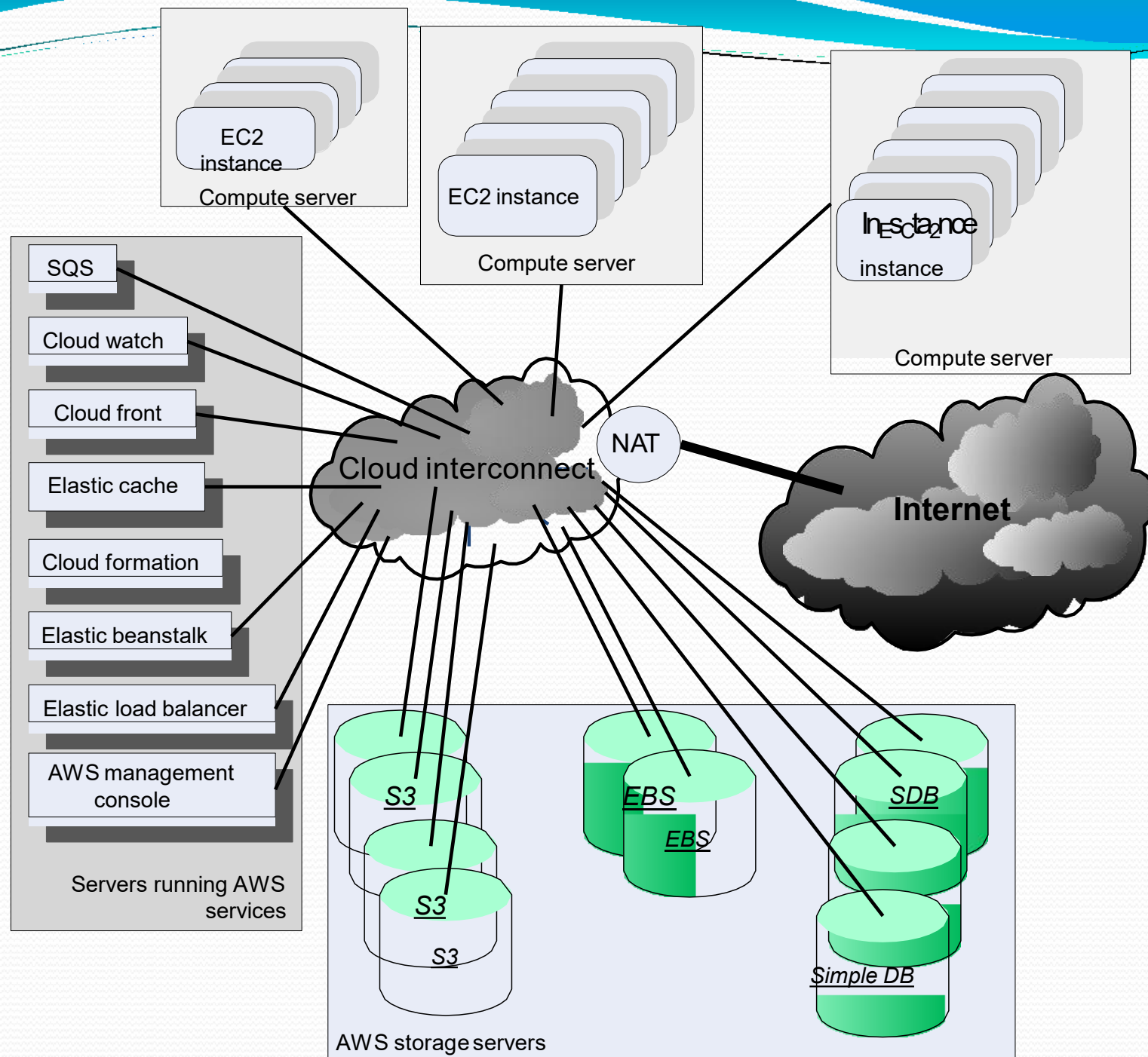


FIGURE 5.1. Amazon infrastructure cloud



Steps to run an application

- Retrieve the user input from the front-end.
- Retrieve the disk image of a VM (Virtual Machine) from a repository.
- Locate a system and requests the VMM (Virtual Machine Monitor) running on that system to setup a VM.
- Invoke the Dynamic Host Configuration Protocol (DHCP) and the IP bridging software to set up MAC and IP addresses for the VM.

User interactions with AWS

- The AWS Management Console. The easiest way to access all services, but not all options may be available.
- AWS SDK libraries and toolkits are provided for several programming languages including Java, PHP, C#, and Objective-C.
- Raw REST requests.

Examples of Amazon Web Services

- *AWS Management Console* - allows users to access the services offered by AWS .
- *Elastic Compute Cloud (EC2)* - allows a user to launch a variety of operating systems.
- *Simple Queuing Service (SQS)* - allows multiple *EC2* instances to communicate with one another.
- *Simple Storage Service (S3), Simple DB, and Elastic Block Storage (EBS)* - storage services.
- *Cloud Watch* - supports performance monitoring.
- *Auto Scaling* - supports elastic resource management.
- *Virtual Private Cloud* - allows direct migration of parallel applications.

EC2 – Elastic Compute Cloud

- *EC2* - web service for launching instances of an application under several operating systems, such as:
 - Several Linux distributions.
 - Microsoft Windows Server 2003 and 2008.
 - OpenSolaris.
 - FreeBSD.
 - NetBSD.
- A user can
 - Load an *EC2* instance with a custom application environment.
 - Manage network's access permissions.
 - Run the image using as many or as few systems as desired.

EC2 (cont'd)

- Import virtual machine (VM) images from the user environment to an instance through *VM import*.
- *EC2* instances boot from an AMI (Amazon Machine Image) digitally signed and stored in *S3*.
- Users can access:
 - Images provided by Amazon.
 - Customize an image and store it in *S3*.
- An *EC2* instance is characterized by the resources it provides:
 - VC (Virtual Computers) – virtual systems running the instance.
 - CU (Compute Units) – measure computing power of each system.
 - Memory.
 - I/O capabilities.

Instance types

- Standard instances: micro (StdM), small (StdS), large (StdL), extra large (StdXL); small is the default.
- High memory instances: high-memory extra large (HmXL), high-memory double extra large (Hm2XL), and high-memory quadruple extra large (Hm4XL).
- High CPU instances: high-CPU extra large (HcpuXL).
- Cluster computing: cluster computing quadruple extra large (Cl4XL).

Instance name	API name	Platform (32/64-bit)	Memory (GB)	Max EC2 compute units	I-memory (GB)	I/O (M/H)
StdM		32 and 64	0.633	1 VC; 2 CUs		
StdS	m1.small	32	1.7	1 VC; 1 CU	160	M
StdL	m1.large	64	7.5	2 VCs; 2 × 2 CUs	85	H
StdXL	m1.xlarge	64	15	4 VCs; 4 × 2 CUs	1,690	H
HmXL	m2.xlarge	64	17.1	2 VCs; 2 × 3.25 CUs	420	M
Hm2XL	m2.2xlarge	64	34.2	4 VCs; 4 × 3.25 CUs	850	H
Hm4XL	m2.4xlarge	64	68.4	8 VCs; 8 × 3.25 CUs	1,690	H
HcpuXL	c1.xlarge	64	7	8 VCs; 8 × 2.5 CUs	1,690	H
Cl4XL	cc1.4xlarge	64	18	33.5 CUs	1,690	H

S3 – Simple Storage Service

- Service designed to store large objects; an application can handle an unlimited number of objects ranging in size from 1 byte to 5 TB.
- An object is stored in a bucket and retrieved via a unique, developer-assigned key; a bucket can be stored in a Region selected by the user.
- Supports a minimal set of functions: write, read, and delete; it does not support primitives to copy, to rename, or to move an object from one bucket to another.
- The object names are global.
- S3 maintains for each object: the name, modification time, an access control list, and up to 4 KB of user-defined metadata.

S3 (cont'd)

- Authentication mechanisms ensure that data is kept secure.
- Objects can be made public, and rights can be granted to other users.
- S3 computes the MD5 of every object written and returns it in a field called ETag.
- A user is expected to compute the MD5 of an object stored or written and compare this with the ETag; if the two values do not match, then the object was corrupted during transmission or storage.

Elastic Block Store (EBS)

- Provides persistent block level storage volumes for use with *EC2* instances; suitable for database applications, file systems, and applications using raw data devices.
- A volume appears to an application as a raw, unformatted and reliable physical disk; the range 1 GB -1 TB.
- An *EC2* instance may mount multiple volumes, but a volume cannot be shared among multiple instances.
- EBS supports the creation of snapshots of the volumes attached to an instance and then uses them to restart the instance.
- The volumes are grouped together in Availability Zones and are automatically replicated in each zone.

SimpleDB

- Non-relational data store. Supports store and query functions traditionally provided only by relational databases.
- Supports high performance Web applications; users can store and query data items via Web services requests.
- Creates multiple geographically distributed copies of each data item.
- It manages automatically:
 - The infrastructure provisioning.
 - Hardware and software maintenance.
 - Replication and indexing of data items.
 - Performance tuning.

SQS - Simple Queue Service

- Hosted message queues are accessed through standard SOAP and Query interfaces.
- Supports automated workflows - *EC2* instances can coordinate by sending and receiving SQS messages.
- Applications using SQS can run independently and asynchronously, and do not need to be developed with the same technologies.
- A received message is “locked” during processing; if processing fails, the lock expires and the message is available again.
- Queue sharing can be restricted by IP address and time-of-day.

CloudWatch

- Monitoring infrastructure used by application developers, users, and system administrators to collect and track metrics important for optimizing the performance of applications and for increasing the efficiency of resource utilization.
- Without installing any software a user can monitor either seven or eight pre-selected metrics and then view graphs and statistics for these metrics.
- When launching an Amazon Machine Image (AMI) the user can start the CloudWatch and specify the type of monitoring:
 - Basic Monitoring - free of charge; collects data at five-minute intervals for up to seven metrics.
 - Detailed Monitoring - subject to charge; collects data at one minute interval.

AWS services introduced in 2012

- Route 53 - low-latency DNS service used to manage user's DNS public records.
- Elastic MapReduce (EMR) - supports processing of large amounts of data using a hosted Hadoop running on EC2.
- Simple Workflow Service (SWF) - supports workflow management; allows scheduling, management of dependencies, and coordination of multiple EC2 instances. Workflow service to orchestrate AWS services, automate business processes, or build serverless applications.
- ElastiCache - enables web applications to retrieve data from a managed in-memory caching system rather than a much slower disk-based database.
- DynamoDB - scalable and low-latency fully managed NoSQL database service.
- Amazon DevPay: Amazon DevPay is a simple-to-use online billing and account management service.

AWS services introduced in 2012 (cont'd)

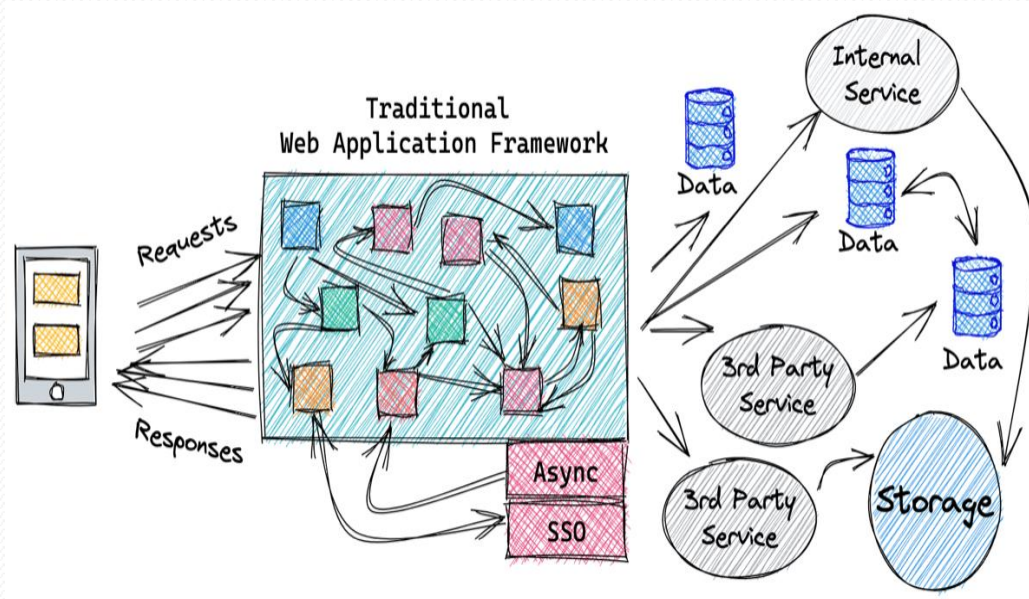
- CloudFront - web service for content delivery.
- Elastic Load Balancer - automatically distributes the incoming requests across multiple instances of the application.
- Elastic Beanstalk - handles automatically deployment, capacity provisioning, load balancing, auto-scaling, and application monitoring functions.
- CloudFormation –
 - allows the creation of a stack describing the infrastructure for an application.
 - AWS CloudFormation gives you an easy way to model a collection of related AWS and third-party resources, provision them quickly and consistently, and manage them throughout their lifecycles.
 - A CloudFormation template describes your desired resources and their dependencies so you can launch and configure them together as a stack.
 - You can use a template to create, update, and delete an entire stack as a single unit, as often as you need to, instead of managing resources individually.
- Amazon Relational Database Service - easy to set up, operate, and scale a relational database in the cloud.

Elastic Beanstalk

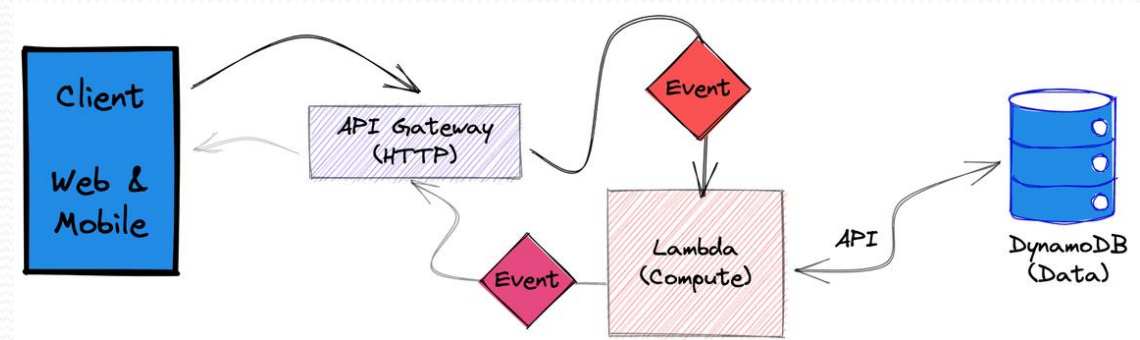
- Handles automatically the deployment, capacity provisioning, load balancing, auto-scaling, and monitoring functions.
- Interacts with other services including *EC2*, *S3*, *SNS*, Elastic Load Balance and AutoScaling.
- The management functions provided by the service are:
 - Deploy a new application version (or rollback to a previous version).
 - Access to the results reported by CloudWatch monitoring service.
 - Email notifications when application status changes or application servers are added or removed.
 - Access to server log files without needing to login to the application servers.
- The service is available using: a Java platform, the PHP server-side description language, or the .NET framework.

What is serverless computing?

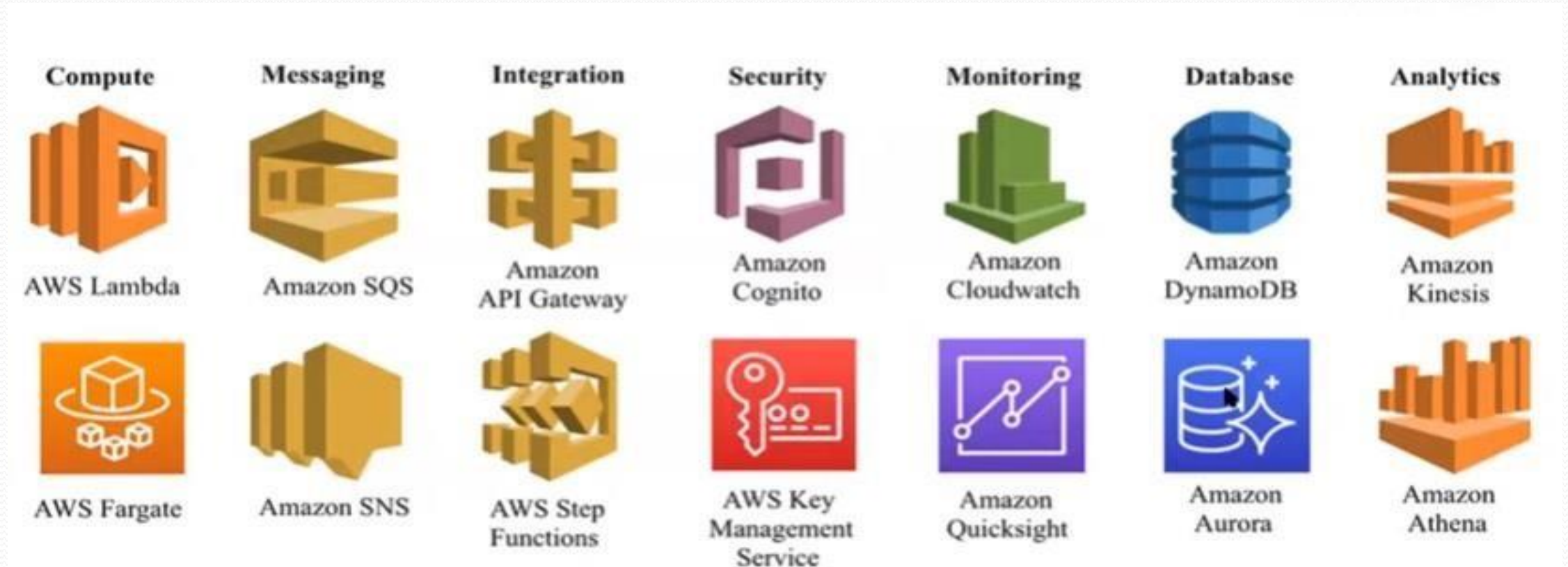
- Serverless computing is a method of providing backend services on an as-used basis.
- A serverless provider allows users to write and deploy code without the hassle of worrying about the underlying infrastructure.
- Your code runs when an event triggers it.
- Increased developer productivity
- Efficient scalability
- Lower costs
- No server management



Event-driven serverless microservice



AWS Serverless Service Stack



AWS Lambda

- AWS Lambda is a serverless compute service that lets you run code without provisioning or managing servers
- Creating workload-aware cluster scaling logic, maintaining event integrations, or managing runtimes
- We can run code for virtually any type of application or backend service - all with zero administration.
- Lambda automatically and precisely allocates compute execution power and runs your code based on the incoming request or event, for any scale of traffic.
- You can write Lambda functions in your favourite language (Node.js, Python, Go, Java, and more) and use both serverless and container tools, such as AWS SAM or Docker CLI, to build, test, and deploy your functions.

AWS Lambda

File Processing



Photograph is taken



Amazon S3
Photo is uploaded
to an S3 Bucket



Lambda is
triggered



AWS Lambda
Lambda runs image
resizing code



Photo is resized into web,
mobile, and tablet sizes

AWS Lambda

Stream Processing



AWS Lambda

Web Application



AWS Lambda

Author from scratch [Info](#)

Name

Runtime

Role
Defines the permissions of your function. Note that new roles may not be available for a few minutes after creation. [Learn more](#) about Lambda execution roles.

Existing role
You may use an existing role with this function. Note that the role must be assumable by Lambda and must have Cloudwatch Logs permissions.

[Cancel](#) [Create function](#)

```
File Edit Find View Goto Tools Window
Environment
  hellowithpyhton
    lambda_function.py
  lambda_function x
    1 def lambda_handler(event, context):
    2     # TODO implement
    3     return 'Hello from Lambda using python'
    4
```

AWS Fargate

- AWS Fargate is a serverless compute engine for containers that works with both Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS).
- Fargate makes it easy for you to focus on building your applications.
- Fargate removes the need to provision and manage servers, lets you specify and pay for resources per application, and improves security through application isolation by design.
- Fargate allocates the right amount of compute, eliminating the need to choose instances and scale cluster capacity.
- You only pay for the resources required to run your containers, so there is no over-provisioning and paying for additional servers.

Amazon Simple Notification Service (Amazon SNS)

- Manage messaging service for both application-to-application (A2A) and application-to-person (A2P) communication.
- The A2A pub/sub functionality provides topics for high-throughput, push-based, many-to-many messaging between distributed systems, microservices, and event-driven serverless applications.
- The A2P functionality enables you to send messages to users at scale via SMS, mobile push, and email.

Amazon API Gateway

- Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale.
- APIs act as the "front door" for applications to access data, business logic, or functionality from your backend services.
- Using API Gateway, you can create RESTful APIs and WebSocket APIs that enable real-time two-way communication applications.
- API Gateway supports containerized and serverless workloads, as well as web applications.
- API Gateway has no minimum fees or startup costs.
- You pay for the API calls you receive and the amount of data transferred out and, with the API Gateway tiered pricing model, you can reduce your cost as your API usage scales.

AWS Step Functions

- AWS Step Functions is a low-code visual workflow service used to orchestrate AWS services, automate business processes, and build serverless applications.
- Workflows manage failures, retries, parallelization, service integrations, and observability so developers can focus on higher-value business logic.

Amazon Cognito

- Amazon Cognito lets you add user sign-up, sign-in, and access control to your web and mobile apps quickly and easily.
- Amazon Cognito scales to millions of users and supports sign-in with social identity providers, such as Apple, Facebook, Google, and Amazon, and enterprise identity providers via SAML 2.0 and OpenID Connect.

AWS Key Management Service

- AWS Key Management Service (KMS) makes it easy for you to create and manage cryptographic keys and control their use across a wide range of AWS services and in your applications.

AWS IAM

- AWS Identity and Access Management (IAM) is a web service that helps you securely control access to AWS resources.
- With IAM, you can centrally manage permissions that control which AWS resources users can access.
- You use IAM to control who is authenticated (signed in) and authorized (has permissions) to use resources.
- When you create an AWS account, you begin with one sign-in identity that has complete access to all AWS services and resources in the account.
- This identity is called the AWS account *root user* and is accessed by signing in with the email address and password that you used to create the account.

Amazon QuickSight

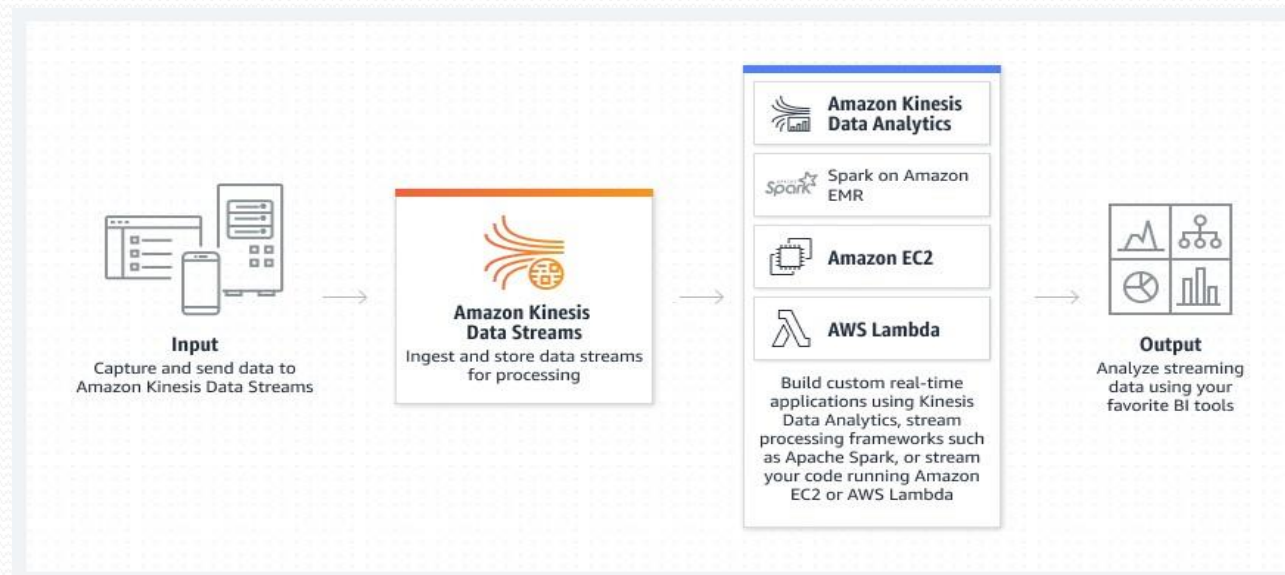
- Amazon QuickSight is a scalable, serverless, embeddable, machine learning-powered business intelligence (BI) service built for the cloud.
- QuickSight lets you easily create and publish interactive BI dashboards that include Machine Learning-powered insights.

Amazon Aurora

- Amazon Aurora is a MySQL and PostgreSQL-compatible relational database built for the cloud, that combines the performance and availability of traditional enterprise databases with the simplicity and cost-effectiveness of open source databases.
- Amazon Aurora is up to five times faster than standard MySQL databases and three times faster than standard PostgreSQL databases.
- Amazon Aurora is fully managed by Amazon Relational Database Service (RDS), which automates time-consuming administration tasks like hardware provisioning, database setup, patching, and backups.

Amazon Kinesis

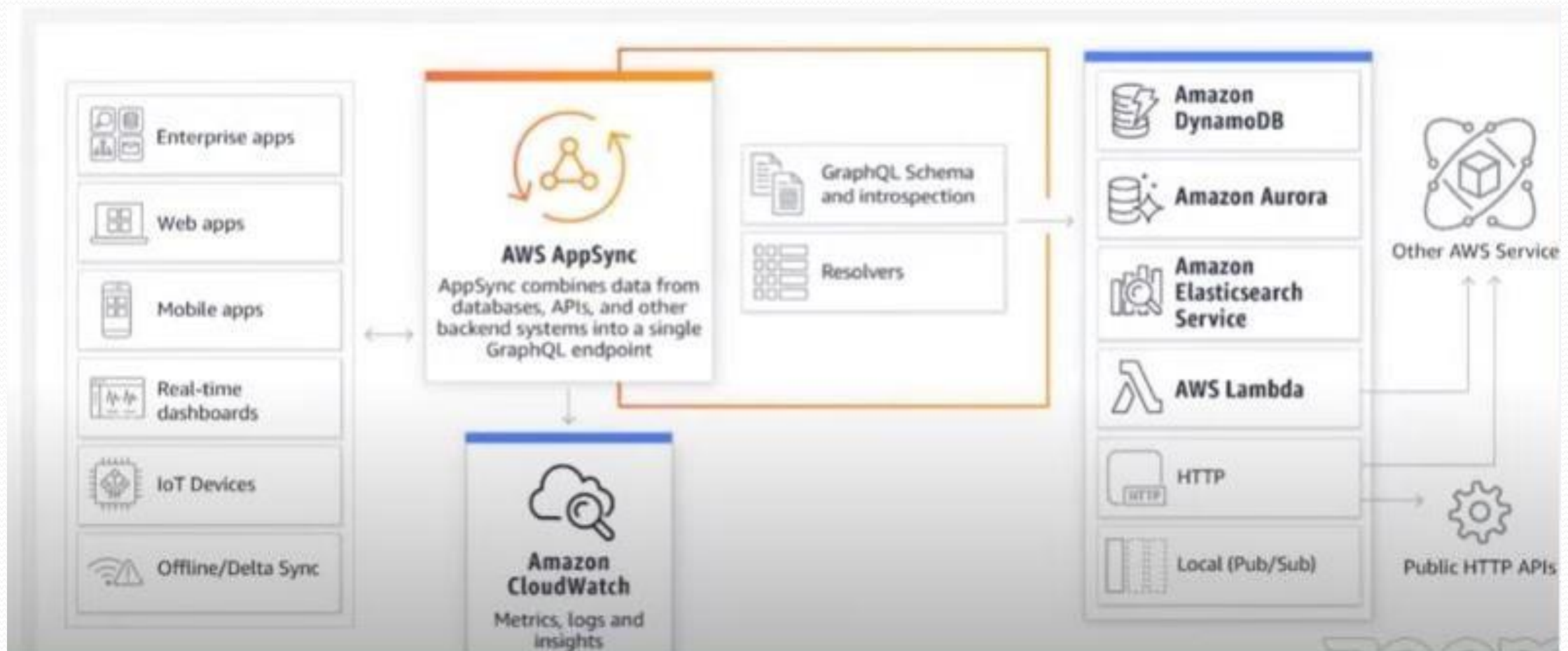
- Amazon Kinesis makes it easy to collect, process, and analyze real-time, streaming data so you can get timely insights and react quickly to new information.
- With Amazon Kinesis, you can ingest real-time data such as video, audio, application logs, website clickstreams, and IoT telemetry data for machine learning, analytics, and other applications.
- Amazon Kinesis enables you to process and analyze data as it arrives and respond instantly instead of having to wait until all your data is collected before the processing can begin.



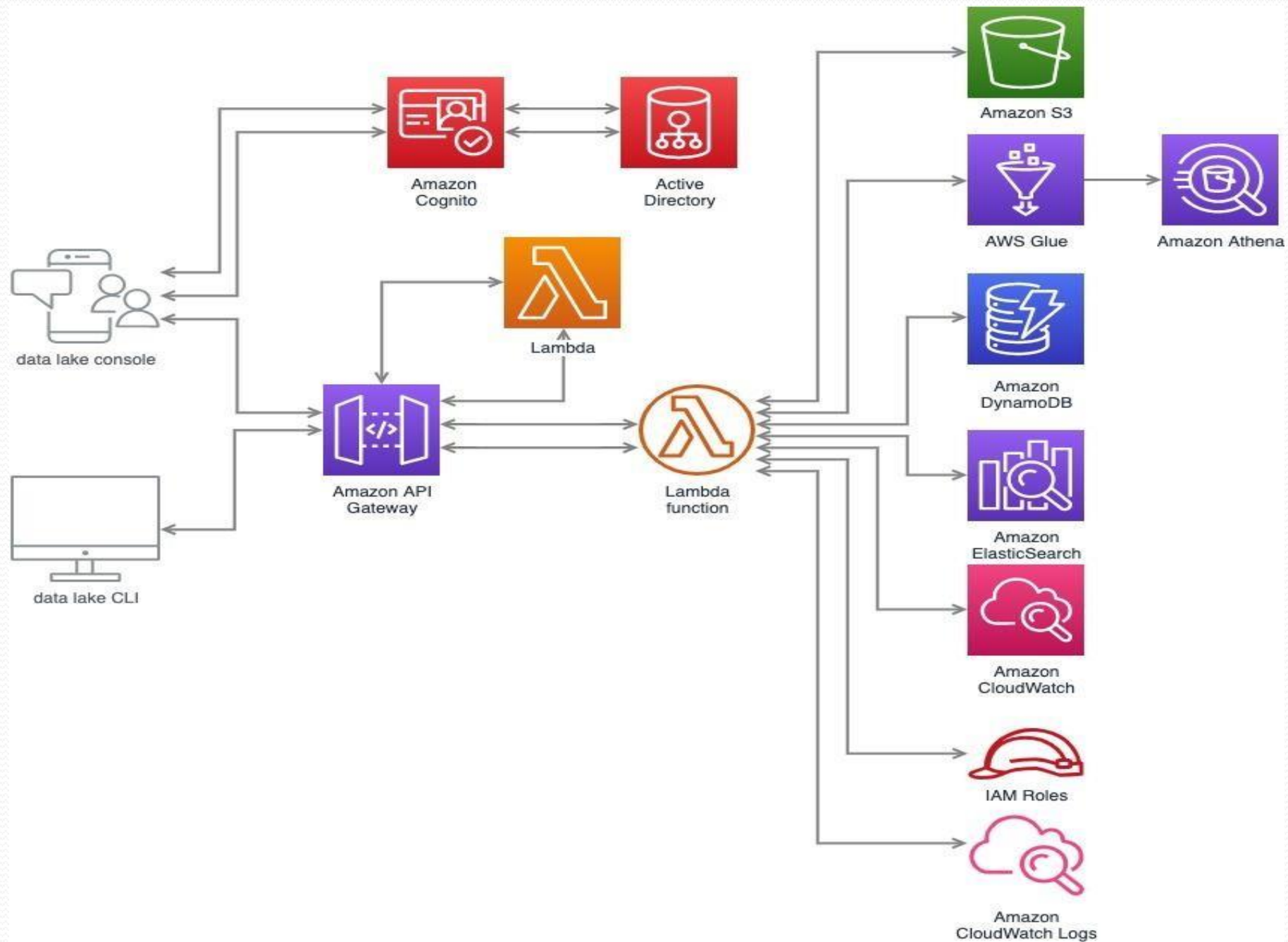
Amazon Athena

- Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 using standard SQL.
- Athena is serverless, so there is no infrastructure to manage, and you pay only for the queries that you run.

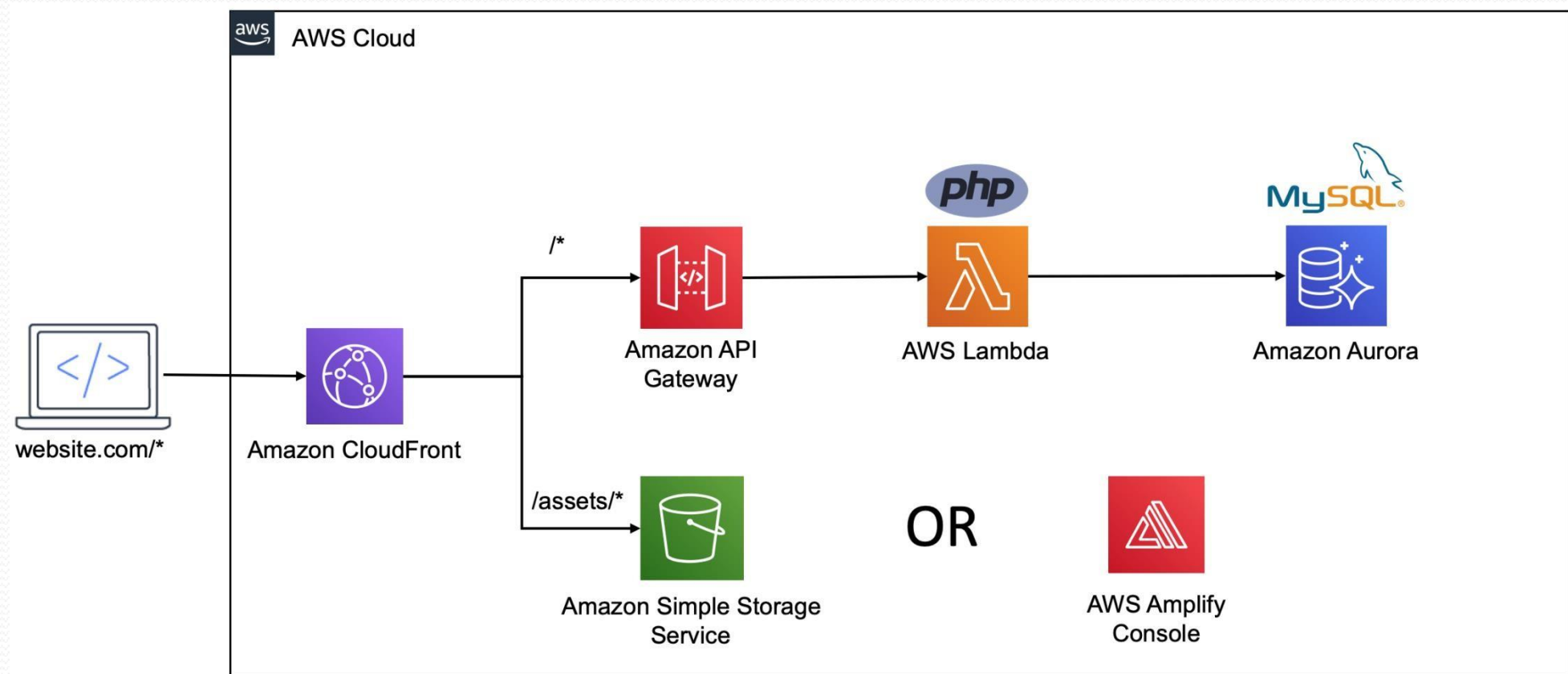
Application



Example



Example



Open-source platforms for private clouds

- Eucalyptus - can be considered as an open-source counterpart of Amazon's EC2.
- Open-Nebula - a private cloud with users actually logging into the head node to access cloud functions. The system is centralized and its default configuration uses the NFS file system.
- Nimbus - a cloud solution for scientific applications based on Globus software, inherits from Globus:
 - The image storage.
 - The credentials for user authentication.
 - The requirement that a running Nimbus process can **ssh** into all compute nodes.

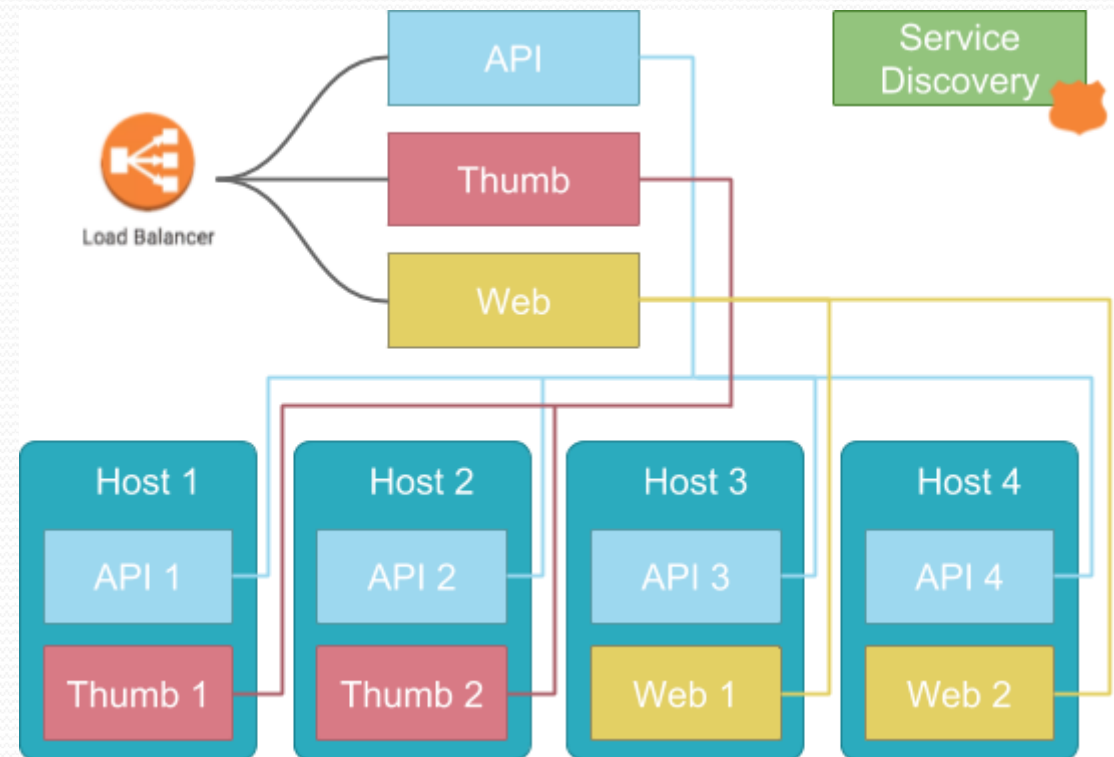
Comparison of Open-source platforms

Table 5: A side-by-side comparison of *Eucalyptus*, *OpenNebula*, and *Nimbus*.

	<i>Eucalyptus</i>	<i>OpenNebula</i>	<i>Nimbus</i>
Design	Emulate EC2	Customizable	Based on Globus
Cloud type	Private	Private	Public/Private
User population	Large	Small	Large
Applications	All	All	Scientific
Customizability	Administrators limited users	Administrators and users	All but image storage and credentials
Internal security	Strict	Loose	Strict
User access	User credentials	User credentials	x509 credentials
Network access	To cluster controller	-	To each compute node

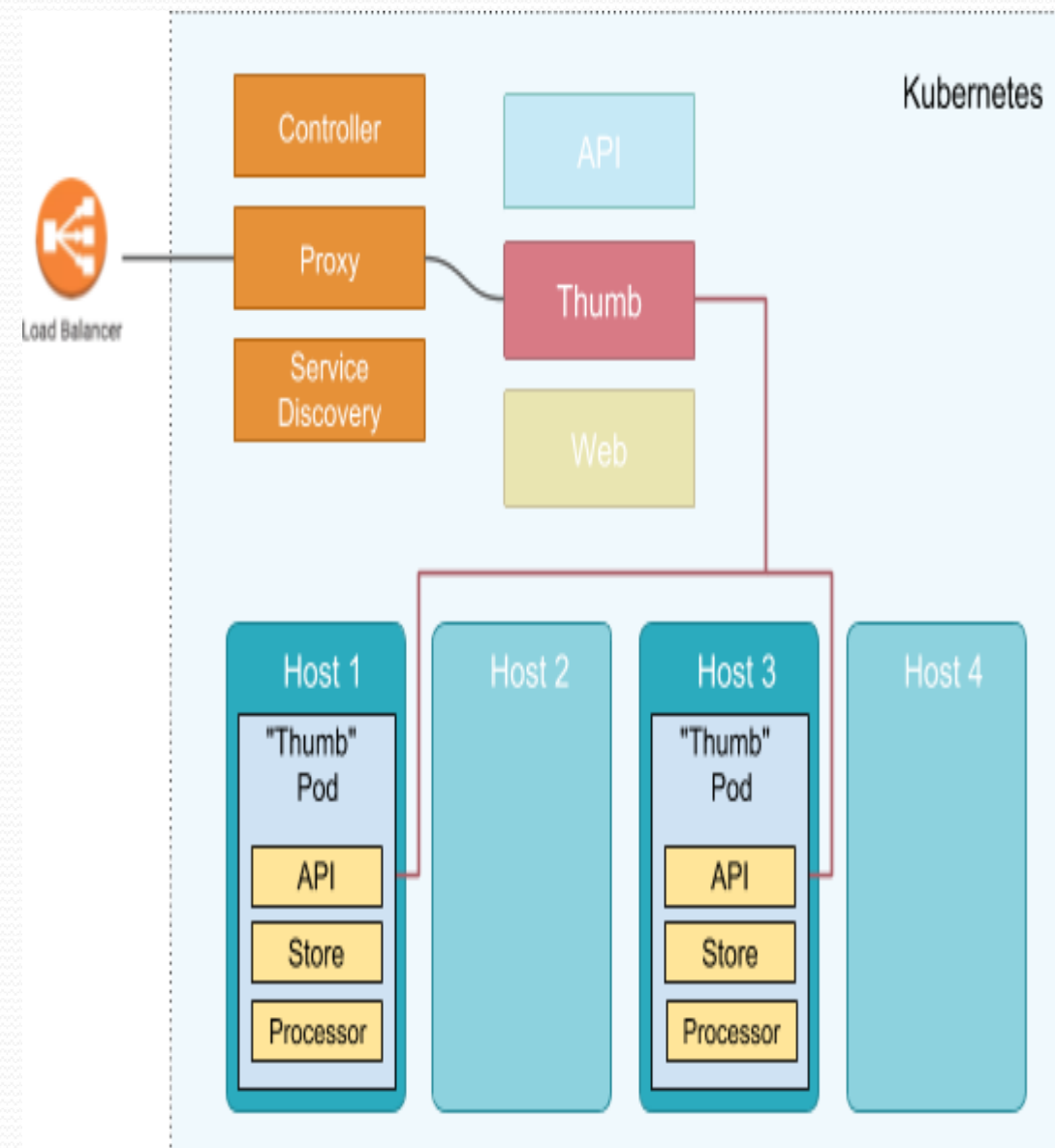
Amazon Elastic Container Service (ECS)

- The Elastic Container Service is a fully managed container orchestration service.
- Amazon Elastic Container Service uses its scheduler to determine:
 - Where a container is run
 - The number of copies started
 - How resources are allocated



Elastic Kubernetes Service (EKS)

- The Elastic Kubernetes Service is essentially a fully managed Kubernetes Cluster. The primary difference between ECS and EKS is how they handle services such as networking and support components.
 - ECS relies on AWS-provided services like ALB, Route 53, etc.,
 - EKS handles all these mechanisms internally, just as in any old Kubernetes cluster.
- Pods can contain either one or more containers with a shared resource pool and provide far more flexibility and fine-grained control over components within a service.



	 Amazon ECS	 Amazon EKS
Open source	No - AWS proprietary	Yes - Kubernetes
Atomic Container Term	Task	Pod
Deployment Effort	Easy (AWS Dashboard)	Medium (AWS plus Kubernetes knowledge required)
Security (IAM)	Comes with the service	Requires addon software and additional configuration
Security (ENI support)	Yes - per task (single container)	Yes - per Pod (which can serve multiple containers)
Per VM Container Limit	Up to 120	Up to 750 Pods (which can host multiple containers)
System Service Cost	Used resources	Used resources plus ~\$75 per cluster per month
Multi-cloud integration	No - AWS specific	Yes - Public and Private cloud integration.

AWS by Category: Core Services

Compute



Amazon
EC2



AWS
Lambda



Auto
Scaling



AWS Elastic
Beanstalk



Amazon Elastic
Container
Registry



Amazon Elastic
Container Service



Amazon
Lightsail



AWS
Batch

Networking



Amazon
VPC



Amazon
Route 53



AWS Direct
Connect



Elastic Load
Balancing

Storage



Amazon S3



Amazon
EBS



Amazon
CloudFront



Amazon
Glacier



Amazon Elastic
File System



AWS
Snowball



Storage
Gateway



AWS
Snowmobile

Database



Amazon
RDS



Amazon
DynamoDB



Amazon
Redshift



AWS
Database
Migration
Service



Amazon
ElastiCache

AWS by Category: Foundational Services

Analytics



Amazon EMR



AWS Data Pipeline



Amazon Elasticsearch



Amazon Kinesis



Amazon Machine Learning



Amazon QuickSight



Amazon Redshift



Amazon Athena

Enterprise Apps



Amazon WorkSpaces



Amazon WorkMail



Amazon WorkDocs

Mobile Services



AWS Mobile Hub



Amazon SNS



Amazon Cognito



AWS Device Farm



Amazon Mobile Analytics



AWS Mobile SDKs



Amazon Pinpoint

Internet of Things



AWS IoT



AWS Greengrass

AWS by Category:

Developer and Operations Services

Developer Tools



AWS
CodeCommit



AWS
CodeDeploy



AWS
CodePipeline



AWS
CodeBuild



AWS X-Ray

Management Tools



Amazon CloudWatch



AWS CloudFormation



AWS
CloudTrail



AWS
Config



AWS
OpsWorks



AWS
Service Catalog



AWS Trusted
Advisor



AWS
Organizations

Security & Identity



AWS
Identity and Access
Management



AWS
Directory Service



Amazon Inspector



AWS CloudHSM



AWS
Key Management
Service



AWS
WAF



AWS Certificate
Manager



AWS
Shield

App Services



Amazon API Gateway



Amazon
AppStream



Amazon CloudSearch



Amazon Elastic
Transcoder



Amazon
SES



Amazon
SNS



Amazon
SQS



Amazon
SWF

Core Services: The Basics

Compute



Amazon
EC2



AWS
Lambda



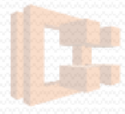
Auto
Scaling



AWS Elastic
Beanstalk



Amazon Elastic
Container
Registry



Amazon Elastic
Container Service



Amazon
Lightsail



AWS
Batch

Networking



Amazon
VPC



Amazon
Route 53



AWS Direct
Connect



Elastic Load
Balancing

Storage



Amazon S3



Amazon
CloudFront



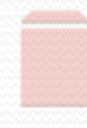
Amazon Elastic
File System



Storage
Gateway



Amazon
Glacier



Amazon
EBS



AWS
Snowball



AWS
Snowmobile

Database



Amazon
RDS



Amazon
DynamoDB



Amazon
Redshift



AWS
Database
Migration
Service



Amazon
ElastiCache

Core Services: The Basics

Management Tools



Amazon CloudWatch



AWS CloudFormation



AWS CloudTrail



AWS Config



AWS OpsWorks



AWS Service Catalog



AWS Trusted Advisor

Security & Identity



AWS Identity and Access Management



AWS Directory Service



Amazon Inspector



AWS CloudHSM



AWS Key Management Service



AWS WAF



AWS Organizations



AWS Certificate Manager



AWS Shield

AWS Machine Learning

PRODUCT	DESCRIPTION
<u>Amazon Polly</u> Text-to-Speech	Amazon Polly is a service that turns text into lifelike speech, allowing you to create applications that talk, and build entirely new categories of speech-enabled products.
<u>Amazon Transcribe</u> Speech-to-Text	Amazon Transcribe uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately.
<u>Amazon SageMaker</u> Build and Deploy Machine Learning Models	Amazon SageMaker is a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy machine learning (ML) models quickly.

AWS Machine Learning

Amazon Textract

Extract Text and Data

Amazon Textract is a machine learning (ML) service that automatically extracts text, handwriting, and data from scanned documents.

Amazon Kendra

Find Accurate Information Faster

Amazon Kendra is an intelligent enterprise search service that helps you search across different content repositories with built-in connectors.

AWS Machine Learning

Amazon Personalize

Personalize Online Experiences

Amazon Personalize enables developers to build applications with the same machine learning (ML) technology used by Amazon.com for real-time personalized recommendations.

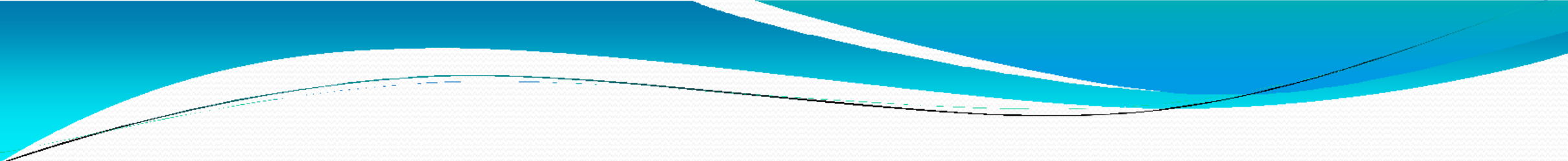
Amazon Rekognition

Automated Image and Video Analysis

Amazon Rekognition makes it easy to add image and video analysis to your applications using proven, highly scalable, deep learning technology that requires no machine learning expertise to use. With Amazon Rekognition, you can identify objects, people, text, scenes, and activities in images and videos, as well as detect any inappropriate content.

AWS Machine Learning

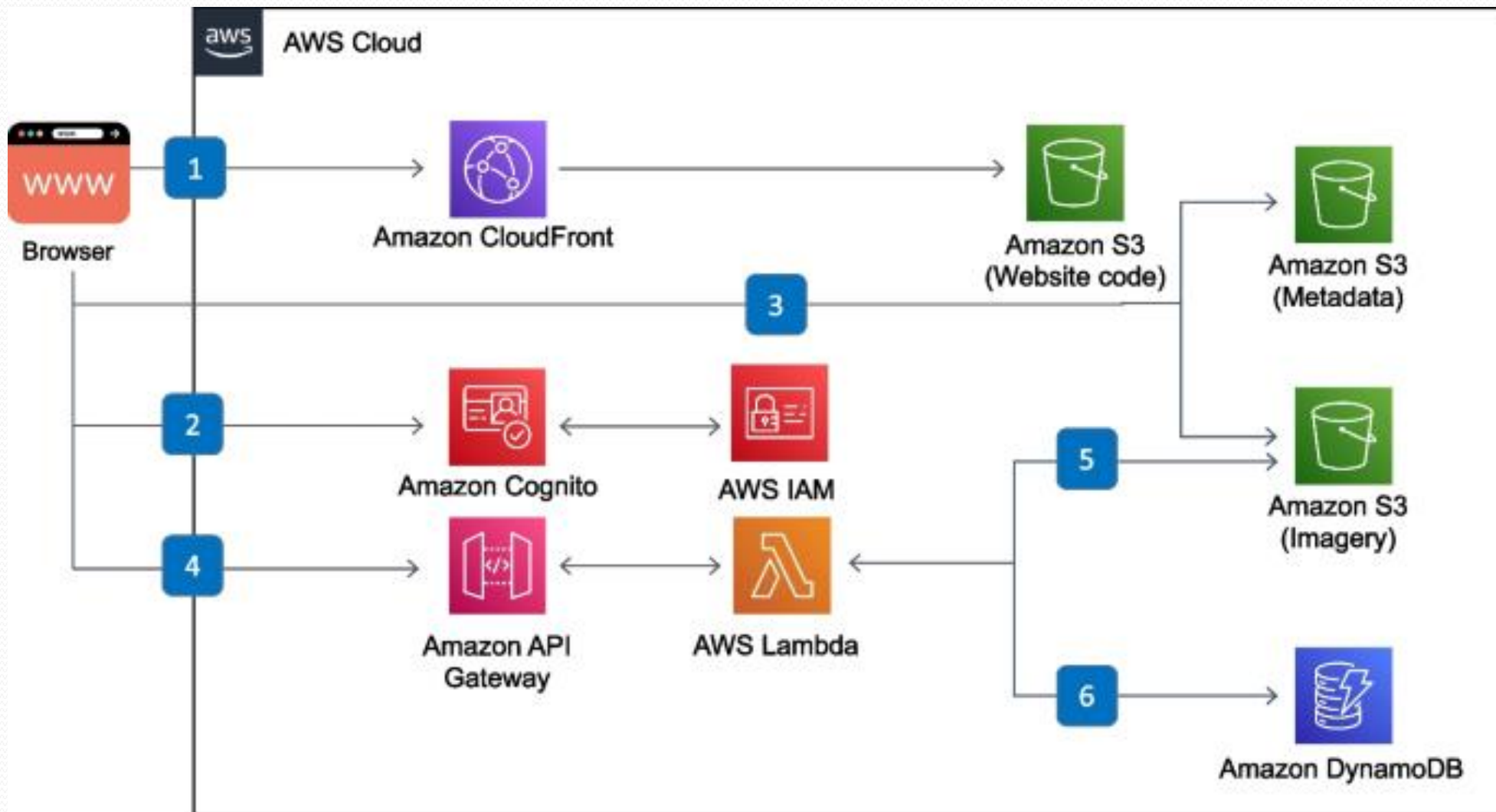
<u>Amazon Lex</u> Conversational AI for Chatbots	Amazon Lex is a service for building conversational interfaces into any application using voice and text.
<u>Amazon Comprehend</u> Natural-Language Processing	Amazon Comprehend is a natural language processing (NLP) service that uses machine learning to find insights and relationships in text. No machine learning experience required.
<u>Amazon Translate</u> Translation	Amazon Translate is a neural machine translation service that delivers fast, high-quality, and affordable language translation.



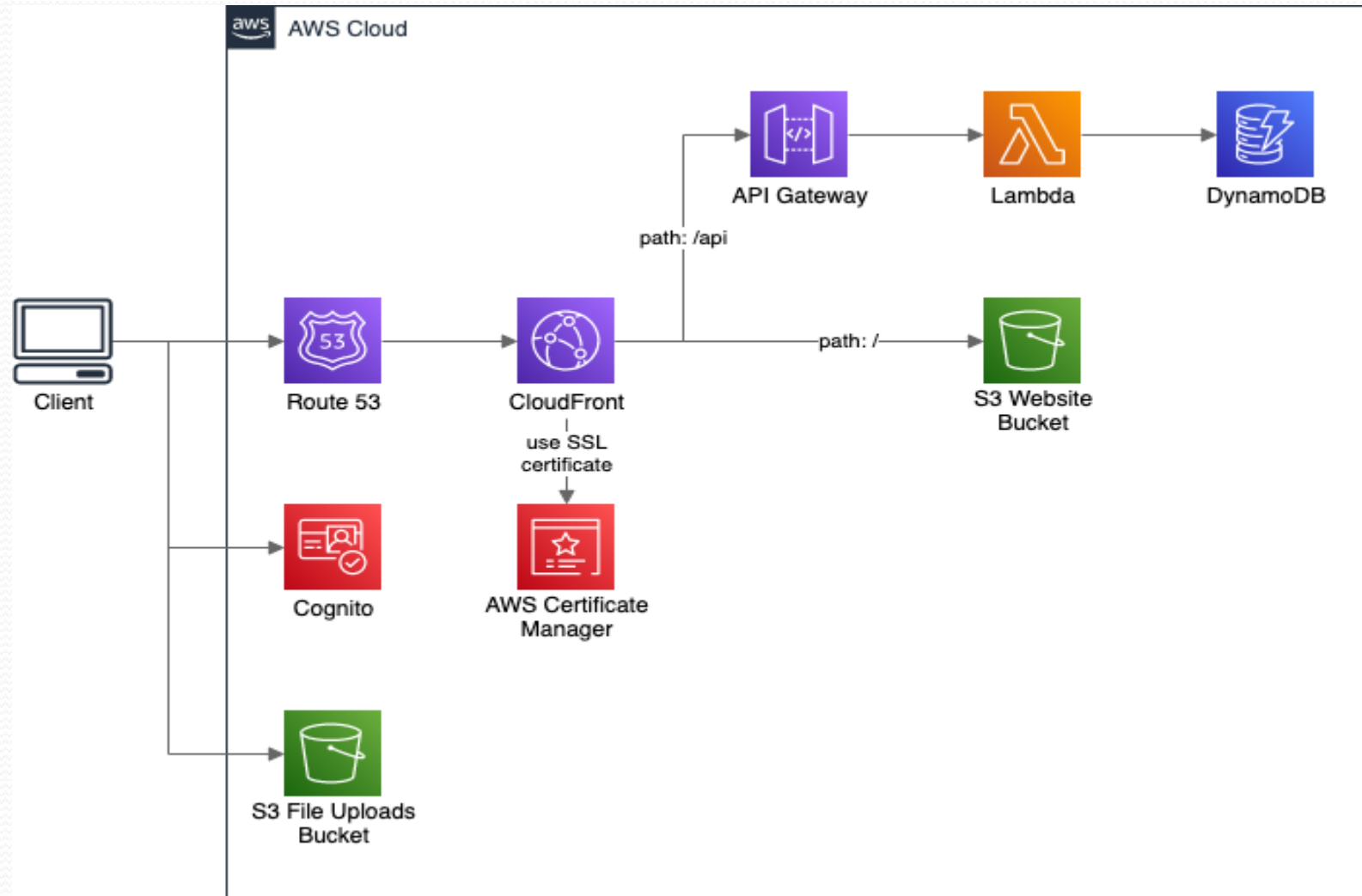
THANK YOU

- An entity tag (ETag) is an HTTP header used for Web cache validation and conditional requests from browsers for resources.
- PEM (**P**rivacy **E**nhanced **M**ail)
- PPK(**P**utty **P**rivate **K**ey) is a windows ssh client,
- It does not support .pem format.
- Hence you have to convert it to .ppk format using PuTTYGen.

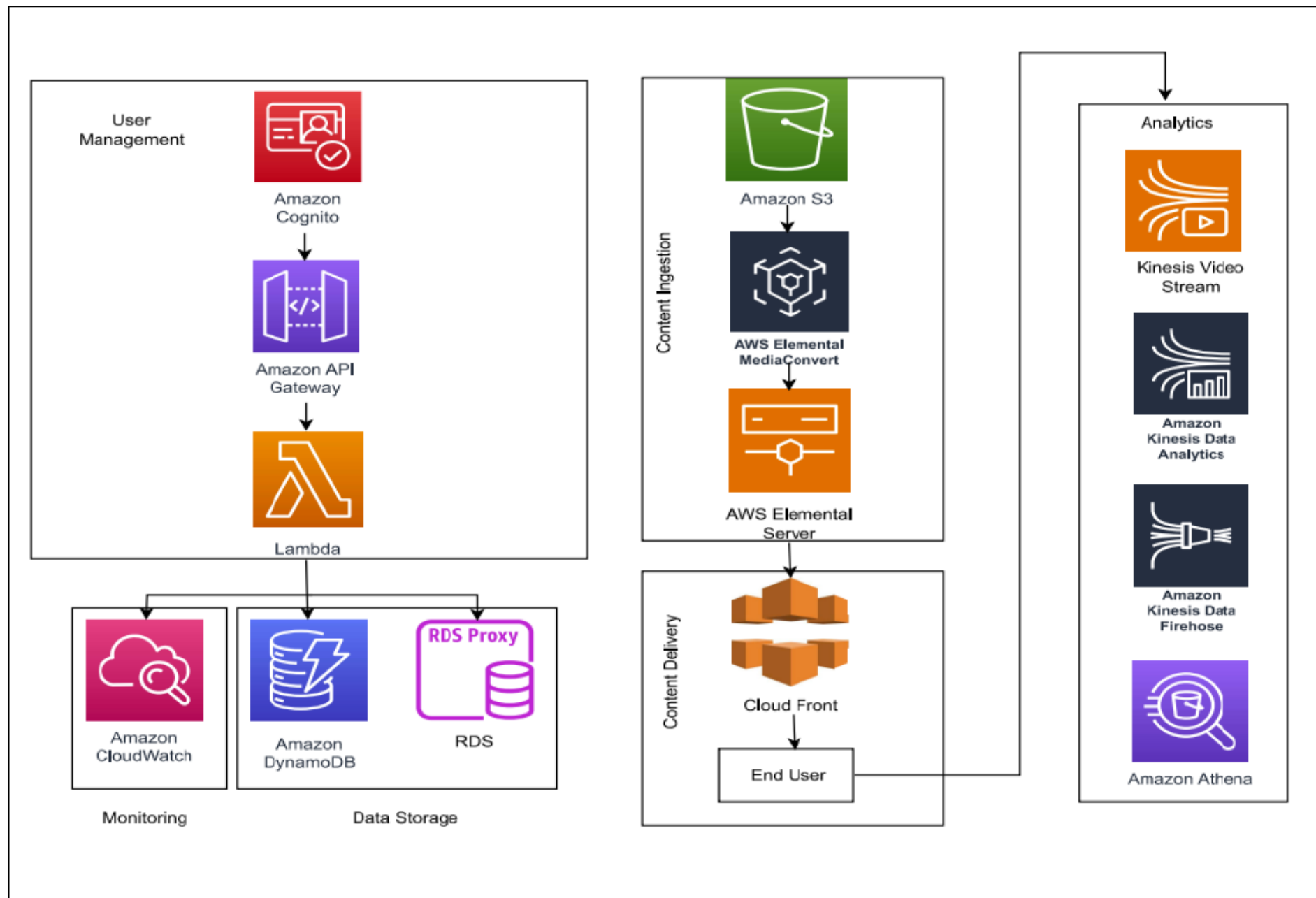
Case Study: Serverless back-end for precomputed result browsing



website hosting on AWS



Video Streaming



Video Streaming

